

Chemical Equilibrium Reactions with Damping Factor

In presence of keywords ***CRDAMP-ALL** and ***CRDAMP**, the chemical reaction terms are computed as:

$$\sigma_{chem,i} = \sum_j v_{ij} \sigma_j \quad (D7.1)$$

where i is the component index, j is reaction index, v is stoichiometry, σ is reaction sink/source term. The reaction term for a particular reaction is computed as

$$\sigma_j = m_{H_2O} k_j \left(1 - \frac{Q_j}{K_{eqj}} \right) \quad (D7.2)$$

where σ_j is the reaction term (gmol/day), m_{H_2O} is the mass of H_2O in kg, k_j is the damping factor in molality/day. The calculated reaction terms Equation (D7.1) are finally added into the component material balance equations.

Keyword ***CRDAMP-ALL** is used to set damping factor for all the chemical reactions, whereas keyword ***CRDAMP** is used to set damping factor for the individual chemical reactions, if necessary.

***REACTION-EQUIL-MNR** (mineral equilibrium reactions) also use Equation (D7.1) and (D7.2).